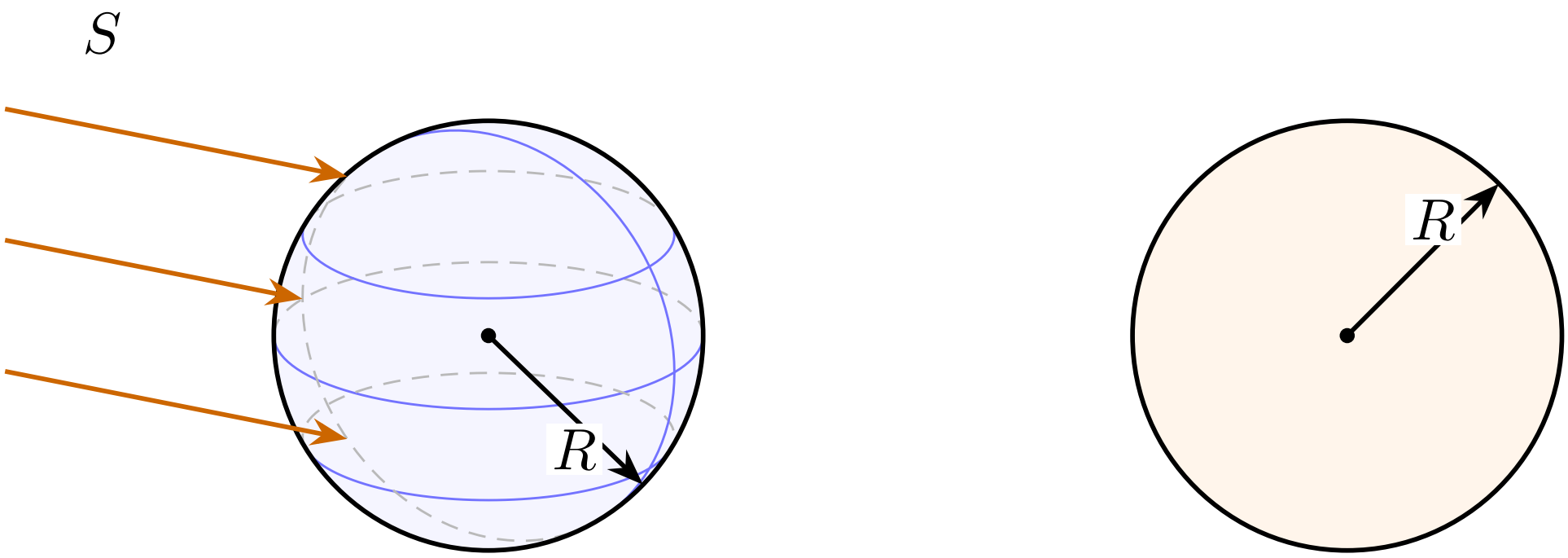
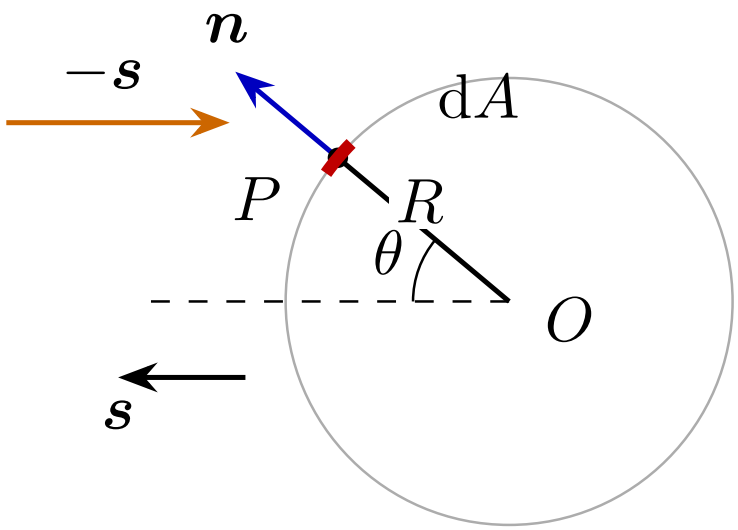
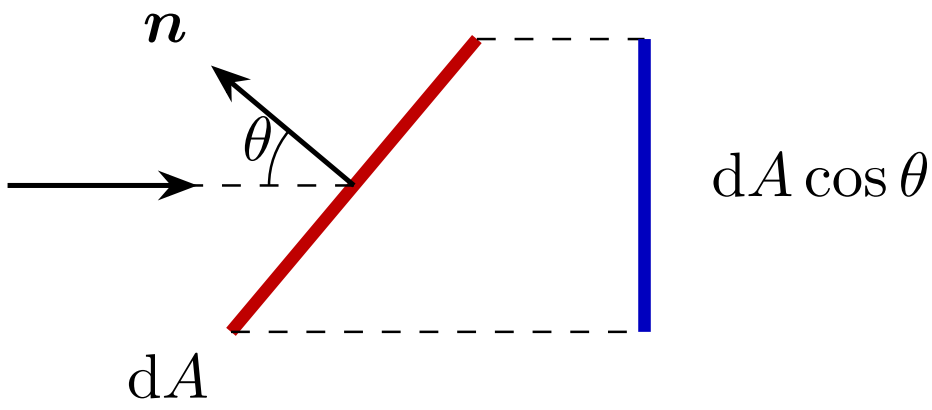


(a) $A_{\text{surf}} = 4\pi R^2$

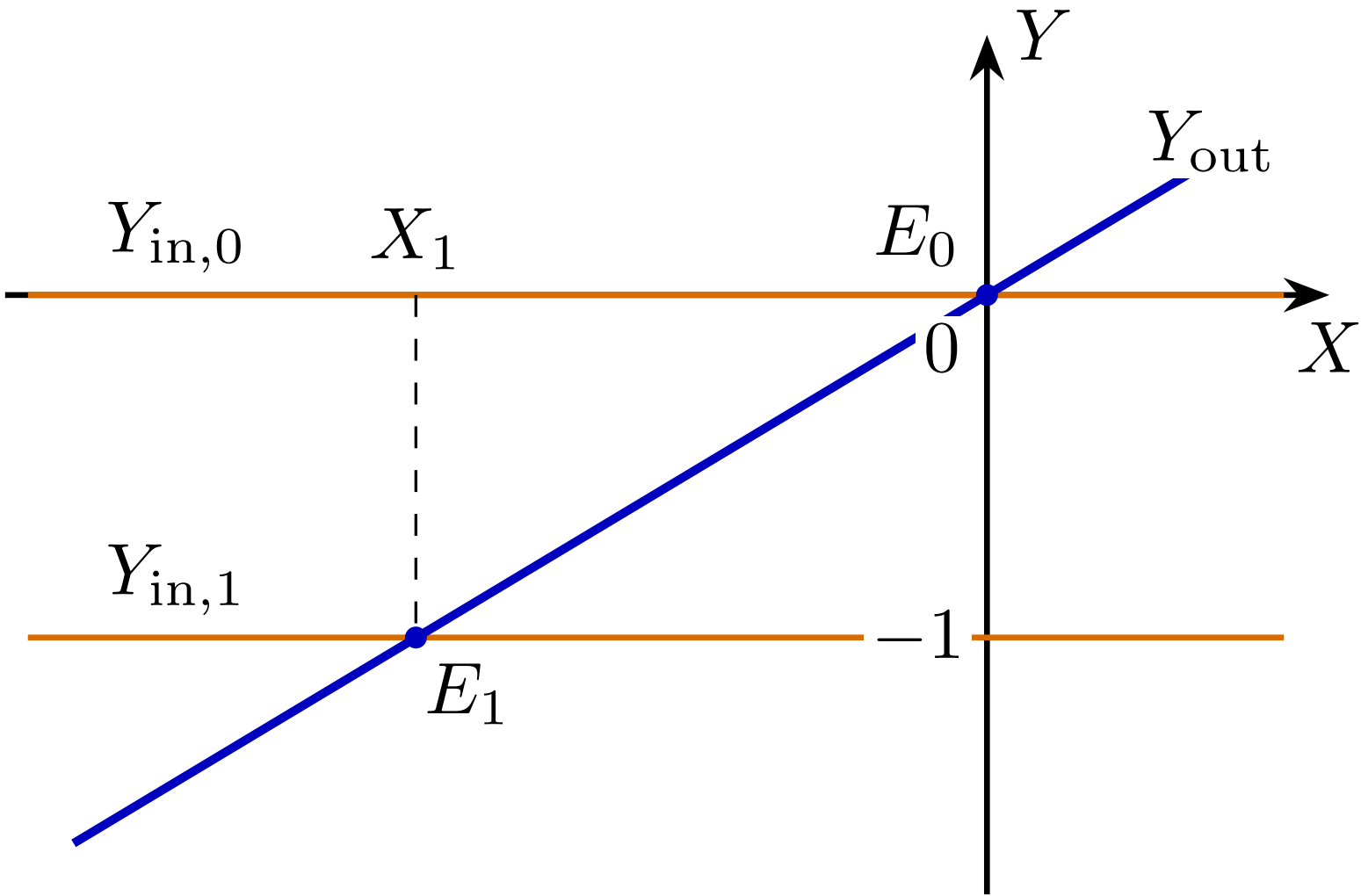
(b) $A_{\text{proj}} = \pi R^2$


$\boldsymbol{n}$
$-\boldsymbol{s}$
$\mathrm{d}A$
P
R
θ
O
$\boldsymbol{s}$

(a)


$\boldsymbol{n}$
θ
$\mathrm{d}A\cos\theta$
$\mathrm{d}A$

(b)


Y
Y_{out}
$Y_{\text{in},0}$
X_1
E_0
0
X
$Y_{\text{in},1}$
-1
E_1